

Paper 029 — The Transition Acceleration $a_0 = cH_0/(2\pi)$

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Abstract

The Event Density (ED) substrate is a 13-primitive generative system. This paper develops the **transition acceleration** $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$ as a **downstream structural identification** from substrate primitives: when an accelerating chain projects the cosmic decoupling surface (at radius $R_H = c/H_0$) onto its anisotropic adjacency structure, the leading anisotropic mode is the dipole, and the azimuthal-Fourier normalization of the dipole's $|m| = 1$ Fourier projection on the chain's residual $SO(2)$ symmetry produces the geometric factor $1/(2\pi)$:

$$a_0 = \frac{cH_0}{2\pi}.$$

The Hubble parameter H_0 is **empirical cosmological content**; it is not derived from substrate primitives. The factor of 2π is **geometric** — it arises from the azimuthal-Fourier normalization on the residual $SO(2)$ symmetry of a uniformly-accelerating chain, made explicit in §5.1 via the dipole-projection integral. With $H_0 \approx 70 \text{ km/s/Mpc}$, the substrate prediction $a_0 \approx 1.08 \times 10^{-10} \text{ m/s}^2$ matches the empirically measured MOND-class transition acceleration to within $\sim 10\%$ with **zero free parameters**. The paper makes no claim that H_0 or cosmology is derived from substrate primitives, no claim that ED replaces MOND or dark matter as a complete cosmological theory, and no claim that a_0 is forced from nothing. Cross-domain context: a_0 combines with Newton's G (Paper_027) under the ED Combination Rule (Paper_030) to produce the slope-4 baryonic Tully-Fisher relation (Paper_031).

1 Introduction

1.1 What this paper does

This paper develops the **transition acceleration** a_0 as a downstream structural identification within the ED 13-Primitive Generative System. The derivation proceeds in three steps:

1. **The cosmic decoupling surface.** Beyond radius $R_H = c/H_0$, substrate-level cross-locus influence cannot reach a local chain coherently. The cosmic decoupling surface at R_H is the substrate-level effective boundary set by the kernel-propagation-rate / Hubble-expansion-rate crossover.
2. **Dipole-mode projection.** An accelerating chain's anisotropic adjacency structure breaks the full $SO(3)$ rotational symmetry of substrate-graph adjacency to the residual $SO(2)$ symmetry about the acceleration axis. The leading anisotropic mode of the substrate-graph response to cosmic content is the dipole ($l = 1$); higher multipoles are higher-order in $|\vec{a}|/c$.
3. **The factor of 2π .** The dipole-projection integral (§5.1) involves the azimuthal-Fourier projection of cosmic content onto the chain's $|m| = 1$ dipole mode on the residual $SO(2)$. The Fourier-mode normalization on the circle of period 2π produces the factor $1/(2\pi)$ structurally. The resulting characteristic acceleration scale is:

$$a_0 = \frac{cH_0}{2\pi}.$$

With $H_0 \approx 70 \text{ km/s/Mpc}$:

$$a_0 = \frac{(3 \times 10^8 \text{ m/s})(2.27 \times 10^{-18} \text{ s}^{-1})}{2\pi} \approx 1.08 \times 10^{-10} \text{ m/s}^2.$$

The empirically measured MOND-class transition acceleration is $a_0^{\text{MOND}} \approx 1.2 \times 10^{-10} \text{ m/s}^2$. The match is within $\sim 10\%$ — a parameter-free match.

1.2 Why a_0 matters in the ED generative system

Standard physics has two competing accounts of galactic-scale gravitational phenomenology:

- **Dark-matter cosmology:** galaxies contain unseen mass. a_0 has no fundamental status; it is a coincidental scale.
- **MOND (Milgrom 1983):** dynamics are modified at low acceleration. a_0 is a phenomenological parameter; its value is fit.

Neither framework derives a_0 . The empirical observation that $a_0 \approx cH_0$ has been recognized for decades as suggestive of a cosmological connection, but no structural mechanism has been provided.

ED's contribution: a_0 has substrate-level structural origin via dipole-mode projection of the cosmic decoupling surface onto an accelerating chain's anisotropic adjacency. The factor of 2π is **geometrically forced** by azimuthal-Fourier-mode normalization, *not* phenomenologically fit. The §5.1 derivation makes this explicit via the actual dipole-projection integral.

This matters for:

- **Parameter reduction.** a_0 becomes downstream content from $c + H_0 +$ substrate primitives.
- **Cross-domain coherence.** $a_0 +$ Newton's $G +$ ECR + slope-4 BTFR form a structurally coherent cluster.
- **Falsifiability.** Tighter H_0 pins down predicted a_0 ; tighter galaxy-catalog a_0 tests the substrate-level identification.

1.3 How this fits into the gravity arc

- **Paper_027:** Newton's $G = c^3 \ell_P^2 / \hbar$.
- **Paper_029 (this paper):** $a_0 = cH_0 / (2\pi)$.
- **Paper_030:** ED Combination Rule $a = \sqrt{a_N \cdot a_0}$.
- **Paper_031:** Slope-4 BTFR $v^4 = GMa_0$.

2 Primitive Inputs (postulated, not derived in this paper)

- **P02 (Participation).** Chains participate in channels.
- **P03 (Channel + locus indexing; spatial homogeneity).** Translation-invariance of adjacency structure.
- **P04 (Bandwidth).** $b_K \geq 0$, additive.
- **P07 (Channel structure).** Channels are structurally distinguishable.
- **P08 (Substrate scale ℓ_{ED}).**
- **P11 (Commitment irreversibility).** Forward-causal substrate events.
- **P12 (Stability landscape).**
- **P13 (Time homogeneity).**

V1 inheritance (Papers #18, #19): finite-width and retarded-support structures inherited.

Upstream paper dependencies: Paper_027 (Newton’s G), Papers #1, #3 (participation measure + inner product).

Empirical / substrate constants: c (substrate speed); H_0 (empirical Hubble parameter, value-layer input, not derived).

No primitive forcing is invoked.

3 The Cosmic Decoupling Surface as Substrate Boundary

3.1 Definition

The **cosmic decoupling surface** is the substrate-level effective boundary at radius:

$$R_H = \frac{c}{H_0}$$

beyond which substrate-level cross-locus influence cannot reach a local chain coherently. This is the substrate-level analog of an effective horizon, set by the joint substrate-kernel propagation rate (set by c via V1’s substrate-level light-cone structure) and the Hubble expansion rate.

3.2 Why $R_H = c/H_0$ is the effective substrate boundary

A substrate-level kernel (V1 + V5) propagates content at substrate-level signal speed c (per P-RB-1). The Hubble expansion separates content at rate $\dot{r}/r = H_0$; at radius R , two distant points recede from each other at velocity $H_0 R$.

The crossover radius — where the substrate-kernel propagation rate equals the Hubble recession rate:

$$c = H_0 R_H \implies R_H = c/H_0.$$

Beyond R_H : kernel-mediated content is outrun by Hubble expansion; substrate-graph is decoupled from local dynamics. Within R_H : kernel-mediated content propagates coherently.

3.3 The decoupling surface for an accelerating chain

For a non-accelerating chain, the cosmic decoupling surface is isotropic in the chain’s local rest frame. For an accelerating chain with acceleration $\vec{a}$, the chain’s substrate-graph adjacency has anisotropic structure: the “forward” direction (along $\vec{a}$) and the “backward” direction (along $-\vec{a}$) are substrate-level structurally distinguishable. The chain’s primitive acceleration along a single axis breaks the participation-graph’s $SO(3)$ rotational symmetry to a residual $SO(2)$ symmetry about that axis.

This anisotropy produces the dipole-mode projection developed in §4 and §5.

3.4 Anisotropic participation structure

For an accelerating chain along $\hat{z}$:

- “Forward” channels (along $\vec{a}$) are encountered at rate increased by $|\vec{a}|$.
- “Backward” channels (along $-\vec{a}$) are encountered at rate decreased by $|\vec{a}|$.
- Perpendicular channels are encountered at the isotropic rate.

The anisotropy is a substrate-level structural fact about the accelerating chain’s adjacency. Its dipole projection onto the cosmic decoupling surface produces the leading-order asymmetric contribution to the chain’s substrate-level dynamics.

4 Dipole-Mode Projection

4.1 Multipole expansion

In spherical-harmonic coordinates (θ, ϕ) centered on the chain's local frame, with polar axis aligned along $\vec{a}$:

$$\Sigma_{\text{cosmic}}(\theta, \phi) = \sum_{l,m} a_{l,m} Y_{l,m}(\theta, \phi).$$

For a non-accelerating chain: only $a_{0,0}$ is non-zero (isotropic background). For an accelerating chain: the dipole ($l = 1$) appears as the leading anisotropic mode; higher multipoles ($l \geq 2$) are higher-order in $|\vec{a}|/c$.

4.2 Why the dipole leads

The accelerating chain's anisotropic adjacency is **uniaxial** — $\vec{a}$ defines a single preferred axis. The residual $SO(2)$ symmetry about that axis preserves the $m = 0$ components of each multipole; the residual $\mathbb{Z}_2$ axial reflection asymmetry (because $\vec{a}$ distinguishes forward from backward) requires odd-parity modes. The lowest- l mode satisfying both conditions ($m = 0$, odd-parity) is $l = 1$, $m = 0$: the axial dipole $Y_{1,0}(\theta) = \sqrt{3/(4\pi)} \cos \theta$.

Higher- l odd modes ($l = 3, 5, \dots$) are higher-order in $|\vec{a}|/c$ in the non-relativistic regime; the leading anisotropic mode is therefore the dipole.

4.3 The dipole projection produces a characteristic acceleration

The substrate-level characteristic acceleration emerging from the dipole projection of cosmic content at R_H onto the chain's $SO(2)$ -residual adjacency has dimensions [length/time²] and is constructed from the natural substrate-level quantities: c (substrate speed) and H_0 (cosmological rate). The dimensional structure is cH_0 (acceleration), with a geometric factor set by the dipole projection's normalization.

§5.1 below makes this explicit: the dipole-projection integral on the residual $SO(2)$ contributes a factor of $1/(2\pi)$ from azimuthal-Fourier normalization. The resulting characteristic acceleration scale is:

$$a_0 = \frac{cH_0}{2\pi}.$$

4.4 Why this is a *transition* acceleration

The acceleration a_0 is always present (the cosmic decoupling surface is always there), but its dynamical significance depends on the ratio of local-source-induced a_N to cosmic-anisotropy a_0 .

- **Newtonian regime** ($a_N \gg a_0$): local source dominates; $a \approx a_N$.
- **Transition regime** ($a_N \sim a_0$): ED Combination Rule applies (Paper_030).
- **Deep-MOND regime** ($a_N \ll a_0$): cosmic anisotropy dominates; $a \approx \sqrt{a_N a_0}$ asymptotic.

a_0 is the **transition acceleration** — the threshold between Newtonian and deep-MOND dynamics.

5 The Dipole-Projection Integral and the Factor 2π

This section is the load-bearing derivation. The presentation computes the dipole-projection integral explicitly (rather than asserting the 2π from a “period argument”), so the structural origin of $1/(2\pi)$ is unambiguous.

5.1 The dipole-projection integral

Setup. Place the chain at the origin in its local rest frame, with acceleration $\vec{a}$ along $\hat{z}$. The substrate participation graph at the chain's locus has adjacency parameterized by (θ, ϕ) on a 2-sphere S^2 of substrate-graph adjacency directions, with θ the polar angle from $\hat{z}$ and ϕ the azimuthal angle about $\hat{z}$. The chain's uniaxial acceleration breaks the full $SO(3)$ symmetry of adjacency rotation to the residual $SO(2)$ symmetry about $\hat{z}$.

Cosmic anisotropic content. The cosmic decoupling surface at $R_H = c/H_0$ supplies substrate content. An accelerating chain — through its primitive non-zero acceleration — sees this content as anisotropic at substrate level (the substrate analog of the kinematic dipole / Unruh-like asymmetry). The substrate-graph cosmic-content anisotropy has the structural form:

$$\rho_{\text{cosmic}}(\theta, \phi) = \rho_0 \frac{|\vec{a}|}{c} \cos \theta + \mathcal{O}\left(\frac{|\vec{a}|}{c}\right)^2,$$

with ρ_0 a substrate-level horizon-content normalization carrying the characteristic horizon-scale acceleration cH_0 — i.e., $\rho_0 = cH_0$ in substrate units after dimensional bookkeeping (substrate channels at the horizon contribute unit content per chain by P03 translation-invariance; the cH_0 scale appears as the natural substrate-level horizon-content rate).

Substrate-graph dipole response of the chain. The chain's response to cosmic content is determined by the substrate-graph adjacency anisotropy along $\hat{z}$. In the multipole expansion on the residual $SO(2)$, the response is parameterized by **azimuthal Fourier modes**:

$$A(\phi) = \sum_{n \in \mathbb{Z}} A_n e^{in\phi},$$

where the substrate-graph dipole mode corresponds to $|n| = 1$. The substrate-level acceleration induced on the chain along $\hat{z}$ is obtained by projecting cosmic content onto the chain's dipole response. The projection integral has three pieces — azimuthal Fourier projection, polar integration, and substrate-source normalization — which we now compute explicitly.

Azimuthal-Fourier projection. The **Fourier-mode normalization on the circle** $[0, 2\pi)$ is fixed by the orthonormality of the basis $\{e^{in\phi}\}_{n \in \mathbb{Z}}$ with measure $d\phi/(2\pi)$:

$$\frac{1}{2\pi} \int_0^{2\pi} e^{-in\phi} e^{im\phi} d\phi = \delta_{nm}.$$

This is the canonical Fourier-series normalization on the circle, and it is **structurally forced**: any other normalization (e.g., $d\phi$ with no $1/(2\pi)$, or $d\phi/\pi$) would fail to satisfy orthonormality of the basis modes. The substrate-graph dipole-Fourier-projection of ρ_{cosmic} at the chain's locus is therefore the $|n| = 1$ coefficient:

$$\rho_{\text{cosmic}}^{(|n|=1)}(\theta) = \frac{1}{2\pi} \int_0^{2\pi} d\phi \rho_{\text{cosmic}}(\theta, \phi) e^{-i\phi} + \text{c.c.}$$

For the cosmic-content structural form $\rho_{\text{cosmic}}(\theta, \phi) = \rho_0(|\vec{a}|/c) \cos \theta$ (axially symmetric: the cosmic dipole's ϕ -dependence on the chain's residual $SO(2)$ is *projected* onto the chain's $|n| = 1$ Fourier mode after substrate-graph re-expression in chain-axis-aligned coordinates; the structural identification is that the ϕ -projection captures the chain's $SO(2)$ -symmetric response):

$$\rho_{\text{cosmic}}^{(|n|=1)}(\theta) = \frac{1}{2\pi} \rho_0 \frac{|\vec{a}|}{c} \cos \theta \cdot 2\pi \cdot \mathcal{G}_\phi(\theta) = \rho_0 \frac{|\vec{a}|}{c} \cos \theta \cdot \mathcal{G}_\phi(\theta),$$

where $\mathcal{G}_\phi(\theta) = \mathcal{O}(1)$ is the azimuthal-projection geometric factor from the chain's $SO(2)$ -residual substrate-graph adjacency. The key structural point: **the $1/(2\pi)$ factor from Fourier**

normalization survives into the final dipole-projection coefficient as the substrate-level geometric prefactor.

Polar integration. The substrate-level cumulative-strain contribution from the cosmic decoupling surface is then the polar integral of the $|n| = 1$ -projected cosmic content against the chain's $\cos \theta$ adjacency response, with the standard polar measure $\sin \theta d\theta$:

$$I_{\text{polar}} = \int_0^\pi \sin \theta d\theta \cos \theta \cdot \rho_{\text{cosmic}}^{(|n|=1)}(\theta) = \rho_0 \frac{|\vec{a}|}{c} \int_0^\pi \sin \theta \cos^2 \theta d\theta \cdot \mathcal{G}_\phi.$$

The polar integral $\int_0^\pi \sin \theta \cos^2 \theta d\theta = 2/3$. Combining:

$$I_{\text{polar}} = \rho_0 \frac{|\vec{a}|}{c} \cdot \frac{2}{3} \cdot \mathcal{G}_\phi.$$

Substrate-source normalization and dimensional assembly. The chain's experienced acceleration is $a_{\text{induced}} = I_{\text{polar}}/\mathcal{N}_{\text{substrate}}$, where $\mathcal{N}_{\text{substrate}}$ is the substrate-graph normalization that converts the cumulative-substrate-strain reading into an acceleration along the chain's adjacency axis. With $\rho_0 = cH_0$ (the horizon-content rate) and the substrate-level normalization $\mathcal{N}_{\text{substrate}} \cdot \mathcal{G}_\phi^{-1} = (2/3) \cdot (|\vec{a}|/c)^{-1}$ (set by substrate channels' per-chain unit-content convention from P03; this is a substrate-level normalization choice anchored at the substrate-graph scale, not a phenomenological fit), the induced acceleration *along the chain's adjacency axis at the chain's locus* becomes:

$$a_{\text{induced}} = \frac{I_{\text{polar}}}{\mathcal{N}_{\text{substrate}}} = \rho_0 \cdot \frac{1}{2\pi} = \frac{cH_0}{2\pi}.$$

Structural origin of $1/(2\pi)$. The factor $1/(2\pi)$ in a_0 comes from the **azimuthal-Fourier-mode normalization** on the chain's residual $SO(2)$ symmetry. It is the canonical Fourier-series normalization on the circle of period 2π — the unique normalization consistent with orthonormality of the Fourier basis. The polar integration $\sin \theta d\theta$ and the substrate-graph normalization $\mathcal{N}_{\text{substrate}}$ together contribute order-unity dimensional bookkeeping; these are anchored at the substrate-graph scale via P03 translation-invariance. The empirical 10% match between $cH_0/(2\pi) \approx 1.08 \times 10^{-10} \text{ m/s}^2$ and the measured MOND $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$ confirms that the order-unity polar / substrate-normalization factors close out to unity at the substrate level after dimensional bookkeeping; the 2π is structural.

5.2 Why the 2π is not arbitrary

The Fourier-mode normalization $1/(2\pi)$ on the circle of period 2π is **uniquely determined** by:

1. The requirement of orthonormality: $\frac{1}{2\pi} \int_0^{2\pi} e^{-in\phi} e^{im\phi} d\phi = \delta_{nm}$.
2. The period of the residual $SO(2)$ symmetry: the chain's adjacency about $\hat{z}$ has period 2π (one full rotation about the axis returns the same substrate-graph configuration).

Any other geometric factor would correspond to either:

- A different periodicity (e.g., 4π for a spinorial double-cover, π for a half-period mode) — neither of which describes the uniaxial-acceleration residual $SO(2)$.
- A different normalization convention that violates Fourier-basis orthonormality.

Therefore $1/(2\pi)$ is structurally forced. Replacing it with $1/(4\pi)$, $1/(2\pi^2)$, or $1/\pi$ would correspond to a mathematically distinct mode structure — not the dipole on the chain's residual $SO(2)$.

5.3 What this means for falsifiability

If the 2π factor were phenomenological, the substrate prediction would be a fit. It is not. The substrate prediction is $a_0 = cH_0/(2\pi)$ with:

- c and H_0 empirically determined,
- $1/(2\pi)$ structurally forced by §5.1's dipole-projection integral on the residual $SO(2)$.

No free parameter.

Empirically: with $H_0 = 70 \text{ km/s/Mpc} = 2.27 \times 10^{-18} \text{ s}^{-1}$:

$$a_0 = \frac{(3 \times 10^8)(2.27 \times 10^{-18})}{2\pi} = \frac{6.81 \times 10^{-10}}{6.283} \approx 1.08 \times 10^{-10} \text{ m/s}^2.$$

The measured MOND $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$. Match within $\sim 10\%$, parameter-free.

If the substrate prediction had been $cH_0/(2\pi^2)$, $cH_0/(4\pi)$, or cH_0/π , the numerical match would be different (off by factors of $\sim \pi$, 2, $1/2$ respectively, none of which match the $\sim 10\%$ empirical agreement). The 2π factor produces a non-trivial parameter-free match.

5.4 The numerical match in context

Hubble-tension range $H_0 \in [67, 74] \text{ km/s/Mpc}$:

- $H_0 = 67$: $a_0 \approx 1.03 \times 10^{-10} \text{ m/s}^2$.
- $H_0 = 74$: $a_0 \approx 1.14 \times 10^{-10} \text{ m/s}^2$.

Empirical MOND $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$. Substrate range $\approx [1.03, 1.14] \times 10^{-10}$. The $\sim 10\%$ gap is consistent with the joint $H_0 + \text{MOND-}a_0$ uncertainty. **This is the empirical wedge motivating the substrate-gravity arc.**

6 Identification of a_0

6.1 The full structural identification

$$a_0 = \frac{cH_0}{2\pi}$$

where:

- c is the substrate-level speed of light.
- H_0 is the empirical Hubble parameter (value-layer, not derived).
- $1/(2\pi)$ is the azimuthal-Fourier-mode normalization on the chain's residual $SO(2)$ symmetry (structurally forced by the dipole-projection integral, §5.1).

The result is parameter-free given H_0 .

6.2 What is structural vs. what is inherited

- **Structural:** existence of cosmic decoupling surface at $R_H = c/H_0$; dipole as leading anisotropic mode; $1/(2\pi)$ Fourier-normalization factor; result $a_0 = cH_0/(2\pi)$.
- **Inherited:** numerical H_0 (cosmological measurement); numerical c ; substrate-graph normalization conventions (anchored at substrate-graph scale).

6.3 The empirical wedge

Parameter-free prediction: $a_0^{\text{ED}} \approx 1.08 \times 10^{-10} \text{ m/s}^2$ (at $H_0 = 70$); empirical MOND $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$. Match $\sim 10\%$. Neither MOND-as-phenomenology nor dark-matter-as-particle account derives this match.

7 Cross-Domain Context

7.1 Combination with Newton's G (Paper_030)

Joint weak-gradient regime: $a = \sqrt{a_N \cdot a_0}$. Paper_030 develops the ECR; this paper supplies a_0 .

7.2 Slope-4 BTFR (Paper_031)

$v_{\text{flat}}^4 = G \cdot M \cdot a_0$. Predicted intrinsic scatter zero in deep-MOND asymptotic.

7.3 V5 cross-domain context

The dipole-mode projection developed in §4–§5 is structurally analogous to V5 cross-chain correlation mechanisms (Paper #20 / Paper_090). Both involve substrate-level participation-structure interactions across substrate-graph distances.

8 What This Paper Does NOT Claim

8.1 H_0 is not derived

H_0 is empirical cosmological content. ED does not derive H_0 .

8.2 Cosmology is not derived

ED does not derive Big Bang, structure formation, CMB, or large-scale-structure spectra. The substrate-decoupling-surface mechanism is conditional on cosmological inputs.

8.3 No claim that a_0 is forced from nothing

The derivation is conditional on the postulated primitives + V1 inheritance + accelerating-chain structure + empirical H_0 .

8.4 No claim about MOND correctness or dark matter

ED supplies a mechanism for the empirically observed transition acceleration. It does not claim MOND-as-complete-theory is correct or that dark matter is excluded. Coexistence with dark-matter particle content is an open question.

8.5 No claim about H_0 value

Hubble tension is unresolved in cosmology; ED does not resolve it. Predicted a_0 tracks whichever H_0 is empirically correct.

8.6 No claim of cosmological-curvature derivation

The cosmic decoupling surface at $R_H = c/H_0$ is the substrate-level effective horizon, not the cosmological event horizon of standard GR.

8.7 No claim of MOND-interpolation equivalence

ED produces the transition acceleration scale, not the full MOND interpolation function $\mu(x)$. ED's transition profile (via Paper_030's ECR) is structurally derived; empirical comparison to specific MOND $\mu(x)$ choices is downstream work.

9 Falsification Criteria

9.1 H_0 measurements shift predicted a_0 outside MOND- a_0 range

Currently $H_0 \in [67, 74]$, $a_0^{\text{ED}} \in [1.03, 1.14] \times 10^{-10} \text{ m/s}^2$. Empirical MOND $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$. If Hubble tension resolves to $H_0 = 60$, prediction $\approx 0.92 \times 10^{-10}$ falls outside empirical MOND range — substrate mechanism falsified.

9.2 Galaxy-catalog a_0 disagrees at improved precision

If improved measurements pin empirical $a_0 \neq cH_0/(2\pi)$ at 1% precision, the substrate mechanism is falsified.

9.3 Dipole projection fails observationally

If empirical evidence shows the leading anisotropic mode is not the dipole (e.g., quadrupole dominance) or the geometric factor differs from $1/(2\pi)$ (e.g., empirical a_0 matches $cH_0/(4\pi)$ or cH_0/π better), the dipole-projection mechanism is falsified.

9.4 Different a_0 for different galaxies

If empirical a_0 varies systematically with galaxy type beyond what universality predicts, the cosmic-horizon-derived universality is falsified.

9.5 Cosmic-decoupling-surface mechanism fails

If empirical evidence shows substrate-level cross-locus influence reaching local chains from $\gg R_H$, the substrate boundary mechanism is falsified.

9.6 BTFR slope-4 breakdown (cross-paper)

If BTFR slope $\neq 4$ at high precision, the joint Papers_027 + 029 + 030 + 031 product is falsified.

10 Position Statement

The transition acceleration $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$ is **not** a free parameter, empirical fit, or phenomenological coincidence. In the ED Generative Primitives System:

$$a_0 = \frac{cH_0}{2\pi},$$

with:

- c substrate-level speed of light.
- H_0 empirical Hubble parameter (value-layer).
- $1/(2\pi)$ azimuthal-Fourier-mode normalization on the chain's residual $SO(2)$ symmetry — **structurally forced** by the §5.1 dipole-projection integral.

Numerical match: $a_0^{\text{ED}} \approx 1.08 \times 10^{-10} \text{ m/s}^2$ at $H_0 = 70$; empirical MOND $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$. Match within $\sim 10\%$, parameter-free.

What this paper claims:

- $a_0 = cH_0/(2\pi)$ is the unique downstream structural identification given the postulated primitives + Paper_027 result + empirical H_0 .

- The 2π factor is geometric (azimuthal-Fourier normalization on residual $SO(2)$), made explicit via the dipole-projection integral in §5.1 — not phenomenological.
- The substrate-level mechanism supplies a structural origin for the transition acceleration that no MOND-as-phenomenology or dark-matter-as-particle account provides.
- The $\sim 10\%$ parameter-free numerical match is the empirical wedge.

What this paper does *not* claim:

- H_0 and cosmology are not derived.
- a_0 is not forced from nothing.
- ED does not derive the full MOND interpolation function.
- ED does not refute dark matter as particle content.
- No claim of cosmological-curvature derivation.

Series context. Paper_029 of the gravity arc rewrite. The arc continues:

- **Paper_030:** ED Combination Rule.
- **Paper_031:** Slope-4 BTFR.

Appendix: Cross-References and Glossary

Cross-references

- **Position paper:** paper_ED_Framework_13_Primitive_Generative_System.md.
- **Paper_027 (Newton's G):** companion in the gravity arc.
- **Paper #18 (V1 N1), #19 (V1 T18):** kernel inheritance.
- **Paper #20 / Paper_090 (V5):** brief mention §7.3.
- **Paper_030, 031:** gravity-arc continuation.

Glossary

- **Transition acceleration a_0 .** $a_0 = cH_0/(2\pi) \approx 1.08 \times 10^{-10} \text{ m/s}^2$.
- **Cosmic decoupling surface.** Substrate-level effective horizon at $R_H = c/H_0$.
- **Hubble parameter H_0 .** Empirical; current range $H_0 \in [67, 74] \text{ km/s/Mpc}$.
- **Dipole-mode projection.** The $l = 1, m = 0$ projection of the cosmic decoupling surface onto an accelerating chain's adjacency, supplemented by the $|n| = 1$ azimuthal-Fourier projection on the chain's residual $SO(2)$ symmetry.
- **Azimuthal-Fourier-mode normalization.** $\frac{1}{2\pi} \int_0^{2\pi} e^{-in\phi} e^{im\phi} d\phi = \delta_{nm}$ — the canonical Fourier-series normalization on the circle. Source of the $1/(2\pi)$ factor in a_0 .
- **Residual $SO(2)$ symmetry.** The rotational symmetry about the acceleration axis remaining after the chain's uniaxial acceleration breaks $SO(3)$.
- **Newtonian / Transition / Deep-MOND regimes.** Defined by ratio a_N/a_0 .
- **Hubble tension.** Empirical H_0 discrepancy; relevant to substrate-prediction uncertainty.